WORKED SOLUTION

STEP III 2025, Question 7

Pure • Inverse Tangent Functions

STEP III 2025, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Everything turns on one fact: $f(x)\,f(-x) = 1$, because $(\sqrt{x^2+1} - x)(\sqrt{x^2+1} + x) = 1$. Together with $f(x) > 0$, this turns $g(x) + g(-x)$ into the familiar $\tan^{-1} t + \tan^{-1}(1/t)$. For (c), $\tan\frac{1}{2}\theta$ is the half-angle tangent, and with $\tan\theta = 1/x$ it simplifies to $f(x)$. Once g and k are known to agree for $x > 0$, the two integrals in (e) are tied together by symmetry.

Solution

Part (i)

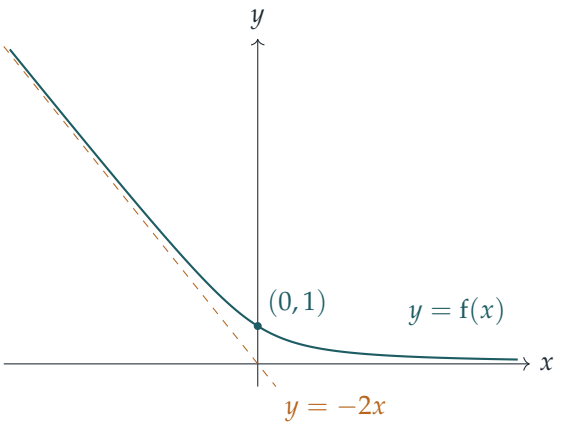
For $x \neq 0$ we have $x^2 = |x|^2$, so

$$\sqrt{x^2+1} = |x|\left(1 + \frac{1}{x^2}\right)^{1/2} = |x|\left(1 + \frac{1}{2x^2} - \frac{1}{8x^4} + \cdots\right) = |x| + \frac{1}{2|x|} - \frac{1}{8|x|^3} + \cdots,$$

using the binomial series, which is valid because $1/x^2 < 1$ when $|x|$ is large. Dropping the terms of order $|x|^{-3}$ gives $\sqrt{x^2+1} \approx |x| + \dfrac{1}{2|x|}$, as required.

The sketch. Gather the facts first.

- $f(x) > 0$ for every x, since $\sqrt{x^2+1} > \sqrt{x^2} = |x| \geq x$. Also $f(0) = 1$.
- $f'(x) = \dfrac{x}{\sqrt{x^2+1}} - 1 < 0$, since $x < \sqrt{x^2+1}$, so f is decreasing. Moreover $f''(x) = (x^2+1)^{-3/2} > 0$, so the curve is convex.
- As $x \to +\infty$: $f(x) \approx x + \frac{1}{2x} - x = \frac{1}{2x} \to 0^+$, so the x-axis is an asymptote.
- As $x \to -\infty$: $|x| = -x$, so $f(x) \approx -2x + \frac{1}{2|x|}$. The line $y = -2x$ is an oblique asymptote, and the curve lies just above it.



Part (ii)(a)

Multiplying out a difference of two squares gives

$$\mathrm{f}(x)\,\mathrm{f}(-x) = \left(\sqrt{x^2+1}-x\right)\left(\sqrt{x^2+1}+x\right) = (x^2+1)-x^2 = 1,$$

so $\mathrm{f}(-x) = 1/\mathrm{f}(x)$. Write $t = \mathrm{f}(x)$, which is positive by part (i). Then

$$\mathrm{g}(x)+\mathrm{g}(-x) = \tan^{-1} t + \tan^{-1}\frac{1}{t}.$$

? **Ask yourself** Why does $\tan^{-1} t + \tan^{-1}(1/t) = \frac{1}{2}\pi$ need $t > 0$?

Let $\alpha = \tan^{-1} t$. Since $t > 0$, $\alpha \in (0, \frac{1}{2}\pi)$, so $\frac{1}{2}\pi - \alpha$ is also in $(0, \frac{1}{2}\pi)$, which lies inside the range of $\tan^{-1}$. Its tangent is $\cot\alpha = 1/t$, so $\tan^{-1}(1/t) = \frac{1}{2}\pi - \alpha$. That gives $\mathrm{g}(x)+\mathrm{g}(-x) = \frac{1}{2}\pi$. (For $t < 0$ the sum would be $-\frac{1}{2}\pi$ instead, which is why the positivity of f matters.)

Part (ii)(b)

$\tan^{-1}$ is an odd function, so for $x \neq 0$,

$$\mathrm{k}(-x) = \tfrac{1}{2}\tan^{-1}\left(-\frac{1}{x}\right) = -\tfrac{1}{2}\tan^{-1}\frac{1}{x} = -\mathrm{k}(x),$$

which means $\mathrm{k}(x)+\mathrm{k}(-x) = 0$.

Part (ii)(c)

Let $x > 0$ and $\theta = \tan^{-1}(1/x)$, so $\theta \in (0, \frac{1}{2}\pi)$ and $\mathrm{k}(x) = \frac{1}{2}\theta$. A right-angled triangle with opposite side 1 and adjacent side x gives $\sin\theta = \dfrac{1}{\sqrt{x^2+1}}$ and $\cos\theta = \dfrac{x}{\sqrt{x^2+1}}$. Using the half-angle identity $\tan\frac{1}{2}\theta = \dfrac{1-\cos\theta}{\sin\theta}$ (from $1-\cos\theta = 2\sin^2\frac{1}{2}\theta$ and $\sin\theta =$

$2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$),

$$\tan \mathrm{k}(x) = \frac{1 - \dfrac{x}{\sqrt{x^2+1}}}{\dfrac{1}{\sqrt{x^2+1}}} = \sqrt{x^2+1} - x = \mathrm{f}(x) = \tan \mathrm{g}(x).$$

This proves the result.

There is more to say. For $x > 0$, $\mathrm{k}(x) \in (0, \frac{1}{4}\pi)$. Also $0 < \mathrm{f}(x) < \mathrm{f}(0) = 1$, so $\mathrm{g}(x) \in (0, \frac{1}{4}\pi)$ too. Since tan is one-to-one on that interval,

$$\mathrm{k}(x) = \mathrm{g}(x) \quad \text{for } x > 0.$$

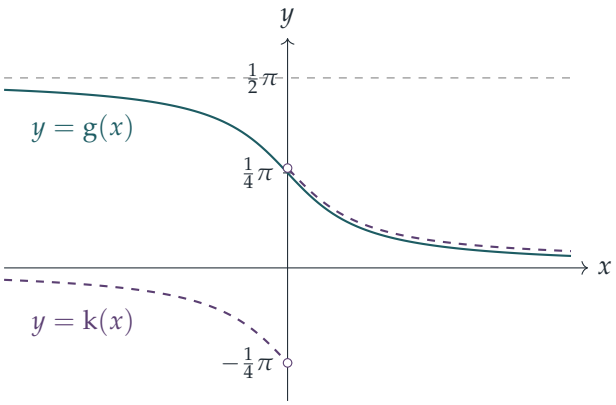
For $x < 0$, we have $-x > 0$, so by (b), then this, then (a):

$$\mathrm{k}(x) = -\mathrm{k}(-x) = -\mathrm{g}(-x) = \mathrm{g}(x) - \tfrac{1}{2}\pi \quad \text{for } x < 0.$$

Part (ii)(d)

$\mathrm{g} = \tan^{-1} \circ \mathrm{f}$ is decreasing, because both $\tan^{-1}$ (increasing) and f (decreasing) are monotonic. As $x \to -\infty$, $\mathrm{f} \to \infty$ and $\mathrm{g} \to \frac{1}{2}\pi$. As $x \to +\infty$, $\mathrm{f} \to 0^+$ and $\mathrm{g} \to 0$. Also $\mathrm{g}(0) = \tan^{-1} 1 = \frac{1}{4}\pi$, and by (a) the graph has rotational symmetry about $(0, \frac{1}{4}\pi)$.

From (c), the graph of k is the same as that of g for $x > 0$, falling from $\frac{1}{4}\pi$ (not attained) towards 0. For $x < 0$ it is the graph of g moved down by $\frac{1}{2}\pi$, falling from 0 (as $x \to -\infty$) towards $-\frac{1}{4}\pi$ as $x \to 0^-$. So k is odd and jumps by $\frac{1}{2}\pi$ at $x = 0$, where it is undefined.



(For $x > 0$ the two curves coincide. The dashed curve is drawn slightly above the solid one there only so that both can be seen.)

Part (ii)(e)

For $x > 0$, $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}\dfrac{1}{x} = \dfrac{-1/x^2}{1+1/x^2} = -\dfrac{1}{1+x^2}$. Integrating by parts, with $\tan^{-1}(1/x)$ differentiated and 1 integrated,

$$\begin{aligned}\int_0^1 \mathrm{k}(x)\,\mathrm{d}x &= \tfrac{1}{2}\Big[x\tan^{-1}\frac{1}{x}\Big]_0^1 + \tfrac{1}{2}\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x \\ &= \tfrac{1}{2}\cdot\tfrac{1}{4}\pi + \tfrac{1}{4}\Big[\ln(1+x^2)\Big]_0^1 = \frac{\pi}{8} + \frac{1}{4}\ln 2.\end{aligned}$$

At the lower limit, $x\tan^{-1}(1/x) \to 0$ as $x \to 0^+$, since $\tan^{-1}(1/x)$ stays below $\frac{1}{2}\pi$. (The integrand is bounded, so the integral is perfectly proper.)

Hence. By (c), $\mathrm{g}(x) = \mathrm{k}(x) + \frac{1}{2}\pi$ for $-1 \le x < 0$. By (b), k is odd, so $\int_{-1}^0 \mathrm{k}(x)\,\mathrm{d}x = -\int_0^1 \mathrm{k}(x)\,\mathrm{d}x$. Therefore

$$\int_{-1}^0 \mathrm{g}(x)\,\mathrm{d}x = \tfrac{1}{2}\pi - \Big(\frac{\pi}{8} + \frac{1}{4}\ln 2\Big) = \frac{3\pi}{8} - \frac{1}{4}\ln 2.$$

Checks

- On $[-1, 0]$, g decreases from $\mathrm{g}(-1) = \tan^{-1}(\sqrt{2}+1) = \frac{3}{8}\pi \approx 1.178$ to $\frac{1}{4}\pi \approx 0.785$, so the integral must lie between these values. Our answer is $\frac{3}{8}\pi - \frac{1}{4}\ln 2 \approx 1.0048$, and numerical quadrature gives the same to ten places. Likewise $\int_0^1 \mathrm{k}\,\mathrm{d}x \approx 0.56599$.
- Differentiating gives $\mathrm{g}'(x) = -\dfrac{1}{2(1+x^2)}$, so in fact $\mathrm{g}(x) = \frac{1}{4}\pi - \frac{1}{2}\tan^{-1}x$. This formula makes (a) obvious, and integrating it directly reproduces the answer to (e).
- At $x = 1$: $\mathrm{f}(1) = \sqrt{2} - 1 = \tan\frac{1}{8}\pi$, and $\mathrm{k}(1) = \frac{1}{2}\cdot\frac{1}{4}\pi = \frac{1}{8}\pi$, as (c) predicts.

Common slips

- Writing $\sqrt{x^2+1} \approx x + \frac{1}{2x}$ for all large $|x|$. For negative x this has the wrong sign, and it hides the asymptote $y = -2x$.
- Concluding $\mathrm{k} = \mathrm{g}$ everywhere from (c). Equal tangents only give equal angles up to multiples of π, and for $x < 0$ the two functions differ by $\frac{1}{2}\pi$.
- Forgetting to justify the limit of $x\tan^{-1}(1/x)$ at 0 in the integration by parts.

Beyond STEP

The function f is $\mathrm{e}^{-\sinh^{-1}x}$, because $\sinh^{-1}x = \ln\big(x + \sqrt{x^2+1}\big)$. Its reciprocal property $\mathrm{f}(-x) = 1/\mathrm{f}(x)$ is just the oddness of $\sinh^{-1}$. The trap in (c), where equal tangents do

not force equal angles, is the real-variable version of the branch problem for inverse functions. It reappears more seriously with complex logarithms and arguments. Needham's *Visual Complex Analysis* is a beautiful place to see this, and Abbott's *Understanding Analysis* treats the limits and improper integrals carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.